DISEASES OF CULTIVATED WILD RICE IN MINNESOTA:

A PICTORIAL GUIDE

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PREFACE

Diseases of Cultivated Wild Rice in Minnesota: A Pictorial Guide is meant to be of use to: farmers; technical representatives; private consultants; county agents; vocational and agricultural instructors and their students in technical, community or 2-year colleges; undergraduate students in agricultural sciences at universities; crop breeders; extension workers; and plant pathologists and graduate students in plant pathology.

The purpose of this guide is to bring together in one easy-to-use book basic information concerning the diseases that affect cultivated wild rice. It is intended to help the user learn more about the diseases of cultivated wild rice -- enabling them to identify the diseases and understand their control.

Each disease is discussed in four parts:

- **Cause.** Causal organism, survival, dissemination and environmental factors favoring development are indicated for each disease.
- **Distribution.** Location the disease is likely to be found.
- **Symptoms.** Visual indicators are provided. Laminated photographs of each disease are included as a hands-on tool to identify disease symptoms.
- **Control.** Measures which can or should be taken once the disease manifests.

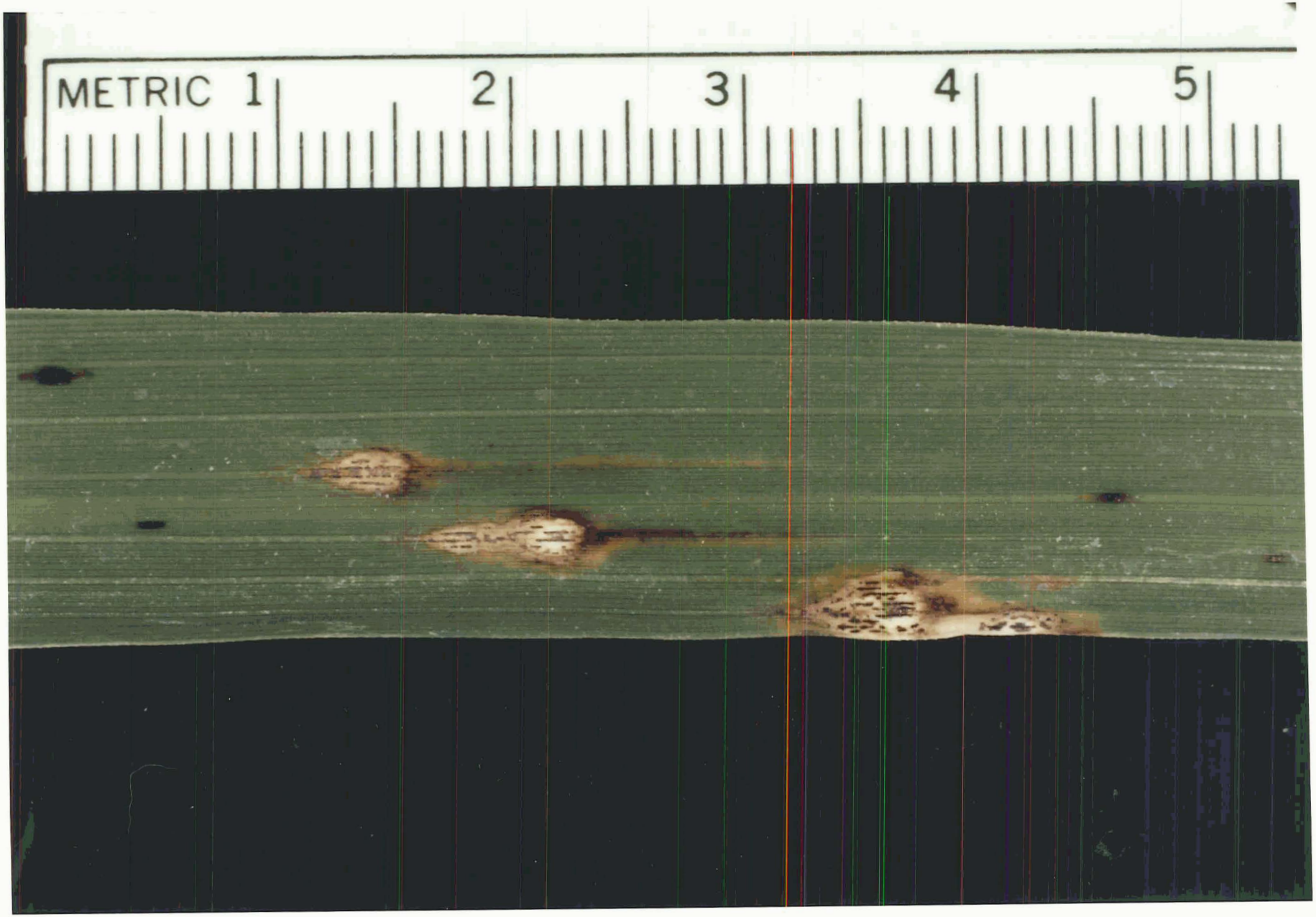
The completeness of a specific disease discussion section is a reflection of our current knowledge and understanding.

Errors of omission or commission are ours and may extend to our choice of words, phrases or contents. For example, where a disease has a causal organism whose name has been changed, the proper name or term to use is discretionary. There is contradictory literature on some diseases and a choice had to be made on what and what not to include in this guide.

This is the first edition of *Diseases of Cultivated Wild Rice in Minnesota: A Pictorial Guide* and the authors would appreciate communication from readers for suggestions or recommendations on how to improve our presentation and also call our attention to errors which may be corrected in the next printing.

JAMES A. PERCICH & ROBERT F. NYVALL

March, 1995


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BROWN SPOT

Cause: *Bipolaris oryzae* (Breda de Haan) Shoemaker. The fungus produces a non-specific toxin called ophiobolin. Studies under controlled conditions indicate infection occurs between 41-97°F with wet periods of 2-36 hours after inoculation. The highest rates of infection can occur at 77-86°F with continuous wet periods of 24 hours. The pathogen can successfully infect and, perhaps, survive on or within the plant residue of all cultivars of wild rice, oats, wheat, corn and at least six species of wild grasses. The conditions necessary for primary infection and dissemination of the pathogen in cultivated wild rice are not well understood.

Distribution: Minnesota natural stands and commercial fields. Epidemics in commercial stands can occur in Minnesota.

Symptoms: All parts of the plant may be infected. If infected the following indicators should be present:

Leaf Symptoms

Tiny brown or purple lesions will be evident first.

Lesions then develop into a characteristic "cigar-shape" with a straw colored-to-light-to-dark brown interior and a yellow halo.

Lesions spread to approximately 1.7 cm (0.5 in.) in length and coalesce.

When brown spot is severe, lesions cover the entire leaf surface causing leaves to wither, shrivel and die.

Stem Symptoms

Stem lesions begin in the same way. Eventually they may girdle the stem, causing it to weaken and break. The head will then fall on the ground or into the water and all seed may be lost.

Flower and Seed Symptoms

Infected flowers and seed become gray in color with spore masses often being present.



ERGOT

Cause: *Claviceps zizaniae* (Fyles) Pantidou. Overwinters as sclerotia. Sclerotia drop in the water and are washed to shore where they survive unfavorable periods. The sclerotia produce stalks on which stroma are borne. Structures (perithecia) in the stroma produce ascospores which are released and become wind-borne to wild rice flowers and infect them. After infection, a sweet, sticky liquid containing conidia is exuded from the flowers. Insects are attracted to the liquid and transmit conidia to healthy flowers. Eventually, the hard sclerotia are formed in place of the grain.

Distribution: Generally distributed wherever wild rice is grown, however, the disease is very rare in commercial fields.

Symptoms: Large ergot bodies or sclerotia replace kernels. Prior to formation of sclerotia, the sticky liquid may be seen on the head together with insects. If infected the following indicators should be present:

First the sclerotia are pink-to-purple when they are growing on the plants, but eventually turn black and very hard when they mature and dry.

Sclerotia vary in size from 3-6 mm (0.12-0.24 in.) in diameter and 3-20 mm (0.12-0.79 in.) in length.

Control: No control other than to remove sclerotia from grain during processing. Chips and pieces of ergot bodies that remain in grain can be floated out with water. Ergot bodies may be harmful if consumed but are larger than kernels and are readily separated from the grain.

Yield loss information is not available.



EYESPOT

Cause: *Drechslera gigantea* (Heald & F. A. Wolf) Ito. All wild rice cultivars are susceptible at all stages of development. The pathogen can infect and survive in the residue of at least 15 different species of crop plants and grasses. Pathogenicity tests indicate that successful infection occurs at 68-95°F with 90-100% relative humidity for at least 48 hours.

Distribution: Generally distributed wherever wild rice is grown, however, the disease is very rare in commercial fields.

Symptoms: If infected the following indicators should be present:

Small, water-soaked, grayish-green lesions on leaves, approximately 1 mm in diameter with well-defined brown margins. Lesions are present by day 5 and sporulation occurs by day 7.

Lesions enlarge becoming tan in the center with secondary lesions producing typical zonate eyespot symptoms. Lesions are 0.8 x 1.5 cm (0.3 x 0.6 in.), although some lesions may be as large as 2.5 x 13 cm (1 x 5 in.).

Under humid conditions, white, prostrate mycelial stands growing from the margins of the lesions may be observed.

Conidia and macroconidia can be found on both surfaces of the infected leaves.

Control: Has not been researched because of infrequent appearance on commercial fields.


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LEAF BLOTCH

Cause: The identity of the fungal pathogen is in doubt, but there are many similarities with the rice pathogen, *Phaeoseptoria oryzae* Miyake. No reliable information exists on survival, infection and dispersal of the leaf blotch pathogen on cultivated wild rice.

Distribution: Reported in natural stands and commercial fields.

Symptoms: Disease of mature and senescent aerial leaves but can be found on floating leaves. Usually begins along the margin or at the base of a leaf blade. If infected the following indicators should be present:

Lesions are initially black, gradually enlarge 1.0 x 2.5 cm (0.4 x 1.0 in.) and interior infected tissue dies and becomes a light tan.

Lesions may cover the entire leaf surface.

Tiny black pycnidia (fruiting structures) develop in the tan regions and produce conidia.

Control: Unknown. Control measures have not been investigated.

Effect unknown.


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LEAF SHEATH AND STEM ROT

Cause: *Nakataea sigmoidea* (Cavara) K. Hara; sclerotial state *Sclerotium oryzae* (Catt.) and *Sclerotium hydrophilum* Sadd.*apud* Rothert. Both pathogens can survive in intact wild rice residue, aquatic weeds, and perhaps, on the soil surface as dormant resting structures called sclerotia. Early in the growing season the disease is most conspicuous as numerous dark lesions on floating leaves. Later in the growing season, lesions caused by *N. sigmoidea* are infrequently found but may be present on all aerial leaves including flag leaves. Dispersal is by sclerotia floating on the water and by airborne conidia. *Sclerotium hydrophilum* dispersal is accomplished by its buoyant, dark brown, more or less spherical sclerotia, which may reach 1 mm in diameter. It is most often associated with a rot of the lower stem and crown. Infection by the sclerotia of either pathogen can also occur on floating leaves and stems.

Distribution: Widespread in both natural stands and commercial fields.

Symptoms: Sclerotia form inside the infected stem and rotted sheaths. If infected the following indicators should be present:

Sclerotia are black, spherical and visible to the unaided eye.

Nakataea sigmoidea have been observed on diseased floating leaves from many locations.

Control: No effective measures to manage these diseases. The leaf sheath and stem rot pathogens survive in wild rice residue which is not incorporated into the soil. However, incorporated residue which does not completely decompose can also be a source of primary infection of floating leaves and stems. Infection studies of the leaf sheath and stem rot pathogens indicate that they do not cause biological or economic crop loss on wild rice. Disease incidence increases with continuous wild rice production. Pathogen association with the insect infestations and secondary bacterial and fungal invaders may cause lodging.



SCAB

Cause: *Fusarium spp.* Fungus overwinter as mycelium and spores on crop residue and other plant hosts. The fungi spread to plants by windborne spores. Infection of wild rice is thought to occur under wet environmental conditions.

Distribution: Unknown. It is thought to be generally distributed wherever wild rice is grown.

Symptoms: Disease spikelets are light brown in color. Other symptoms are not well known.

Control: Not know.



SMUT

Cause: *Entyloma lineatum* (Cke.) Davis and Fisher; synonym (*E. oryzae* on rice). Fungus overwinters on diseased leaf and stem tissue. Spores of *E. lineatum* germinate and infect at 82-86°F. Dispersal is by wind and rain.

Distribution: Generally found wherever wild rice is grown.

Symptoms: Usually do not appear until plants are almost mature. If infected the following indicators should be present:

Glossy black lesions occur on stems, culms and heads. The black discoloration is due to chlamydospores being produced under the surface of the stems and leaves.

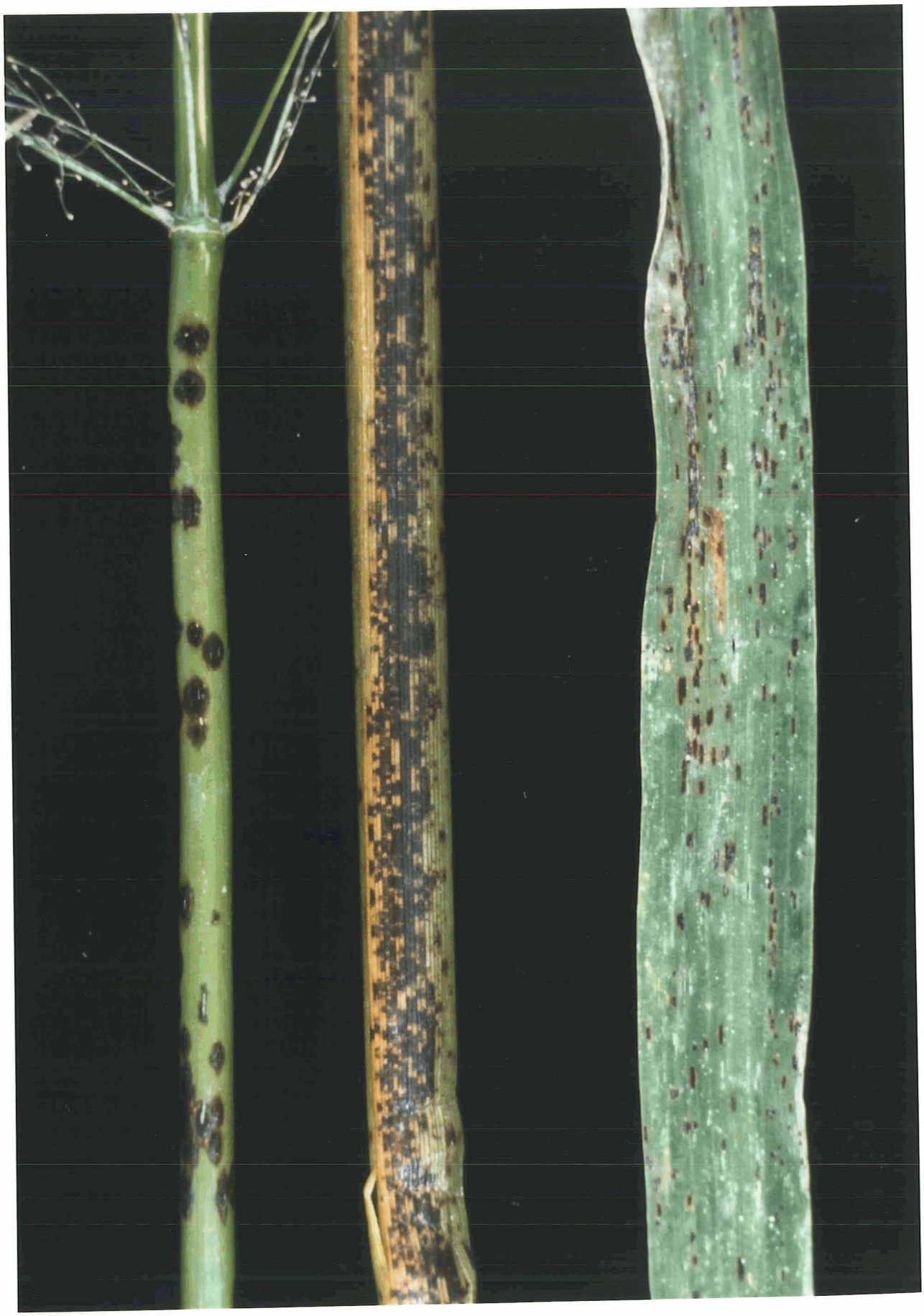
As plants mature and dry, the black color changes to gray.

Lesions can elongate, coalesce and girdle the stems near the head. There is no evidence the disease kills immature plants.

As the disease progresses, lesions may affect all parts of the inflorescence.

Control: No control measure available nor apparently needed. Resistance may have been observed among individual wild rice breeding lines and selections.

Small and insignificant yield reductions may occur.



SPOT BLOTCH

Cause: *Bipolaris sorokiniana* (Sacc.) Shoemaker. Controlled studies of *B. sorokiniana's* overwintering, early infection, dispersal, survival and losses on wild rice are not well understood.

Since the first disease report in 1961, fungal brown spot of cultivated wild rice had been considered as one disease. However, the disease is caused by two related fungi. *Bipolaris oryzae* and *B. sorokiniana* that produce two different disease symptoms. *Bipolaris oryzae* produces the toxin ophiobolin but *B. sorokiniana* does not. *Bipolaris sorokiniana* survives in grasses and residue and causes disease earlier in the growing season while *B. oryzae* is thought to survive only in plant residue but may be a better parasite and causes disease later in the season. Under optimum conditions for disease, *B. oryzae* predominates but *B. sorokiniana* is the dominant pathogen under less optimum conditions.

Distribution: Widespread in both natural stands and commercial fields.

Symptoms: If infected the following indicators should be present:

Lesions caused by *B. sorokiniana* immediately after infection are small, circular and dark-brown in color.

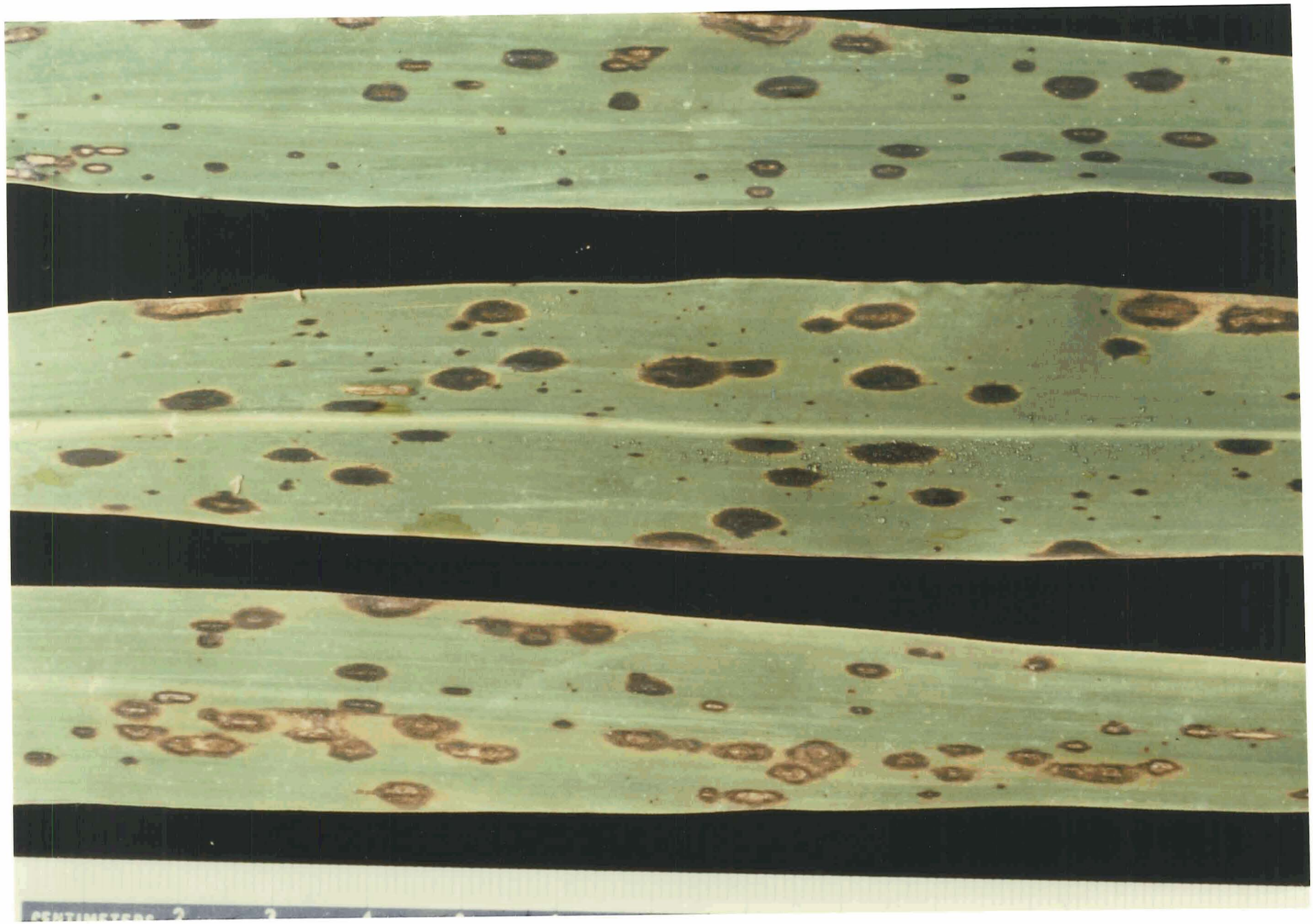
As lesions develop they become large, irregular and may grow together.

Mature lesions are 0.5 x 1.5 cm (0.2 x 0.6 in.) or larger and can be found on floating leaves and all aerial parts of wild rice.

Control: Same as control measures for fungal brown spot since both diseases often occur together. Studies of the pathogen's ability to become resistant to fungicides have been reported.

Crop loss assessments have not been made.

We suggest the name fungal brown spot be retained for the disease caused by *B. oryzae* and spot blotch be used for the disease caused by *B. sorokiniana*.



BACTERIAL BROWN SPOT

Cause: *Pseudomonas syringae* pv. nov. *zizaniae* (Bowden & Percich). Overwintering of the pathogen has not been carefully studied in wild rice fields. The organism has been recovered from soil 3 months after harvest, however, soil is not considered an important source of infection because the bacterium dies during the survival period. The organism is easily recovered from shattered seed, leaf and stem crop residue; and from numerous grass and weed hosts that inhabit dikes and ditches. The pathogen can infect rotational crops such as oats, barley, wheat, rye and corn. Plant wounding is an important means of infection. Disease development occurs at 70-86°F during periods of wetness caused by rain and/or dew and can occur throughout the growing season. All cultivars of wild rice are susceptible to BBS. BBS isolations since 1980 indicate that in some fields, *P. syringae* was the dominant brown spot pathogen and may have affected more than 5% of the leaf area.

Distribution: Generally found wherever wild rice is grown.

Symptoms: If infected the following indicators should be present:

Small water-soaked spot or short streak on the leaf appears first.

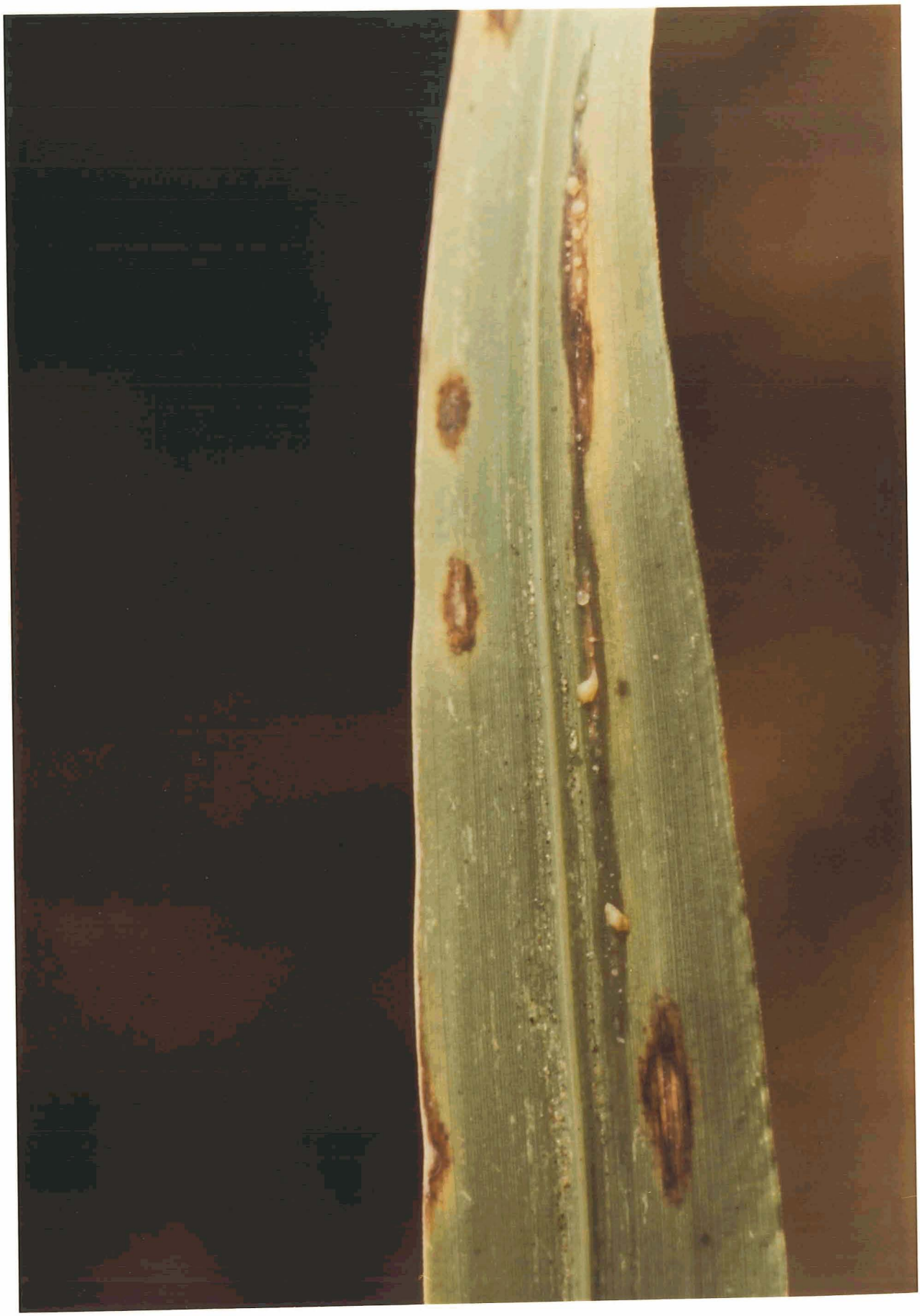
Diffuse greenish-brown halo with a dark brown margin rapidly develops around the central water-soaked area. Greenish-brown area becomes chestnut or tan. Small or narrow lesions appear uniformly dark brown.

Typical lesions are elliptical or spindle-shaped and 2-12 mm long by 1-10 mm wide. Lesions are often irregularly shaped or diffuse and as long as 25 cm (10 in.) or more.

White exudate (bacteria) may be present in center of lesions. In older lesions, translucent areas may fall out, giving a shot hole appearance.

A problem in evaluating the importance of BBS is the difficulty in distinguishing it from spot blotch caused by *Bipolaris sorokiniana*. Careful observation, however, usually reveals a translucent slit typical of BBS. Also, BBS lesions are often irregular and have associated areas of diffuse necrosis (dead tissue), whereas lesions caused by *B. sorokiniana* spp. are typically more uniformly elliptical and have well-defined margins.

Control: No control measures.



BACTERIAL LEAF STREAK

Cause: Bacterial leaf streak of wild rice is caused by at least two different species of bacteria. Both *Xanthomonas campestris* and *Pseudomonas syringae* pv. *zizaniae* pv. nov. can be isolated from infected plants. *Pseudomonas syringae* pv. *zizaniae* can be found 20 times more frequently than *X. campestris* from commercial fields. The mode of infection by the two bacteria are very similar. All wild rice varieties and numerous grasses are susceptible to both pathogens.

Refer to the bacterial brown spot disease description section in this book for details concerning host range, survival, infection and spread of *P. syringae* pv. *zizaniae* and *X. campestris*.

Xanthomonas campestris can also infect wheat, smooth brome and quack grass, barley, rye, timothy and all cultivars of wild rice.

Distribution: Generally found wherever wild rice is grown.

Symptoms: If infected the following indicators should be present:

Translucent, water-soaked spots and streaks, usually 1-3 mm wide.

Lesions may expand lengthwise along the leaf blade up to 25 cm (10 in.), but lateral expansion is restricted by leaf veins.

Water-soaked tissues become necrotic after 1-7 days, giving leaves a brown or black-striped appearance.

Frequently, white flakes (*P. syringae*) or yellow (*X. campestris*) droplets of bacterial exudate are present on the lesion surfaces.

Control: No control procedures are recommended.



WHEAT STREAK MOSAIC VIRUS

Cause: Wheat Streak Mosaic Virus (WSMV). The eriophyid mite vector, *Aceria tulipea* Keif., which is commonly found on wild rice, retains WSMV for several days in its midgut, body cavity and salivary glands. The mite can be transported long distances by wind. The mite can also transmit five other known viruses. Virus-vector relationships are difficult to study because of the small size of the mite. There is no good evidence for replication of viruses in the mite. However, monitoring for WSMV in wild rice is continuing.

Attempts to transmit the virus by green peach aphids (*Myzus persicae* Sulzer) and oat bird cherry aphids (*Rhopalosiphum padi* L.) to wild rice were unsuccessful.

Distribution: Generally distributed wherever wild rice is grown, however, the disease is very rare in commercial fields.

Symptoms: If infected the following indicators should be present:

Typical streak symptoms appear first, but as the disease progresses the chlorotic areas on the infected leaves become necrotic after 10 days.

Necrotic areas eventually coalesce which results in the death of the leaf.

Control: No control measures are recommended at this time.

Although very few wild rice plants were found to be naturally infected with WSMV, crop losses may occur in the future. This virus has caused significant yield losses in other cereals.



JOHNSON BROWN

Cause: Unknown physiological disease.

Distribution: Cultivated wild rice containing Johnson cultivar genetic background.

Symptoms: If infected the following indicators should be present:

Brown-to-black leaf and stem lesions. Rapidly enlarge and coalesce.

Severely affected plants become necrotic and do not produce seed.

Plants inoculated with *Bipolaris oryzae, B. sorokiniana, Drechslera gigantea, Pseudomonas syringae* pv. *zizaniae*, or *Xanthomonas campestris* will result in the symptoms as previously described. The plant appears to have the inability to defend itself against these pathogens.

Control: No control measures.



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BEST MANAGEMENT PRACTICES FOR CULTIVATED WILD RICE



A Guide to Profitable and Environmentally Sound Production

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UNIVERSITY OF MINNESOTA

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Best Management Practices for Cultivated Wild Rice University of Minnesota and Wild Rice Growers

Introduction

The goal of crop producers and environmentalists are the same - to have an abundant and nutritious food supply produced in an environmentally safe manner. Best management practices (BMP's) for wild rice means managing inputs so as to maximize farm profitability while encouraging positive interaction with soil, water and wildlife. They can be more broadly defined as "economically sound, voluntary practices that are capable of minimizing nutrient and chemical contamination of surface and groundwater". Wild rice production requires relatively flat land with a relatively impermeable layer of soil in the upper profile to limit water percolation so water can be maintained on the field during the season. Thus, soil erosion and leaching of nutrients generally are not a problem in wild rice production as they are for upland crop production. However, the aquatic environment requires special care especially during chemical application and drainage before harvest. Best management practices during this time are especially important.

The BMP's described in this publication are based upon research, particularly at the University of Minnesota, other public institutions and private groups, plus practical experiences of wild rice growers. This ensures that the BMP's are technically sound and likely to be easily adopted by growers.

Record keeping is a valuable part of BMP's. It is important when a chemical, for example, is applied so the proper water management can be implemented after treatment. Records of nutrient applications to a particular field are valuable so the correct rates are applied to avoid over-applications. A record of the kinds of chemicals applied is important to avoid improper ones being used. Records of other field operations should also be kept to aid in culture control of weeds in particular.

The BMP's described in this publication are guidelines for wild rice (*Zizania palustris* L.) growers and should be used to enhance the environment and can directly increase efficiency and increase profits. BMP's benefit the grower and the local and the broader society as a whole by enhancing water quality.

Understanding Crop Growth and Development

Efficient wild rice production requires knowledge of the growth and development of the wild rice plant. The stage of plant development is important in the application of fertilizers and pesticides. Fertilizer is more efficient when applied at the right stage, thus often requiring less amounts. The same is true for pesticide applications.

Wild rice growth and development follows other small grains except that the early stages occur underwater, and the third or fourth leaf, depending on water depth, floats on the water surface. The stages of development are germination, emergence, floating leaf, aerial leaf, tillering, jointing, boot, flowering, grain formation and maturity. Figure 1 shows the various growth stages with their corresponding date of development and the growing degree days needed for development.

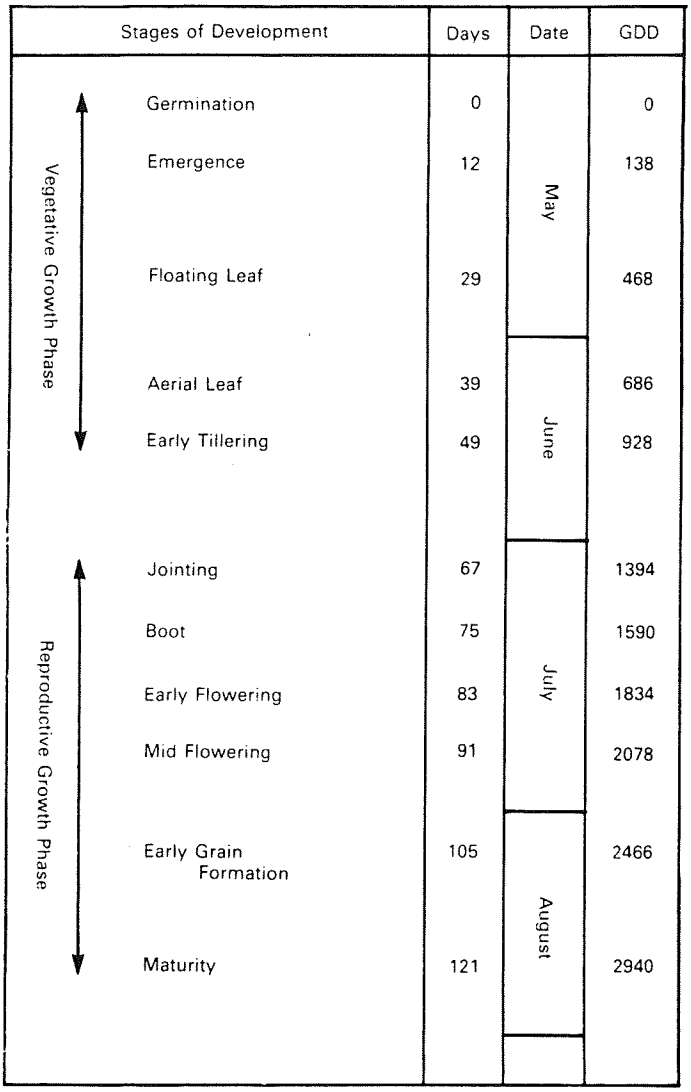


Figure 1. Wild rice plant development (K2 variety) in Aitkin County. GDD = growing degree days using 40° F as the base temperature

Figure 2 shows three stages of wild rice seedlings before emerging from the water surface. In deeper water, the leaf blades elongate more than in shallow water, thus usually the third leaf is the floating leaf and the fourth leaf is the first one to become erect out of the water.

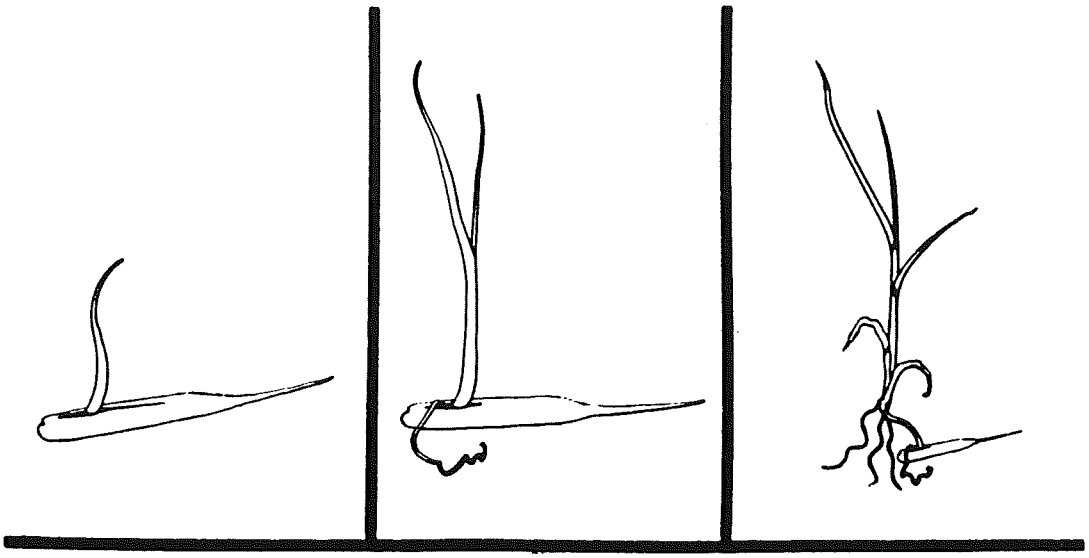


Figure 2. Submerged wild rice seedlings 6, 10, and 18 days old from left to right. The left seedling has the coleoptile emerged, the center seedling has the first true leaf emerged while the left seedling has the coleoptile at the base, three true leaves fully developed and the fourth leaf emerging.

The floating leaf stage is unique to wild rice compared to regular rice (*Oryza sativa* L.). Figure 3 shows a plant in the floating leaf stage and one in the aerial leaf stage.

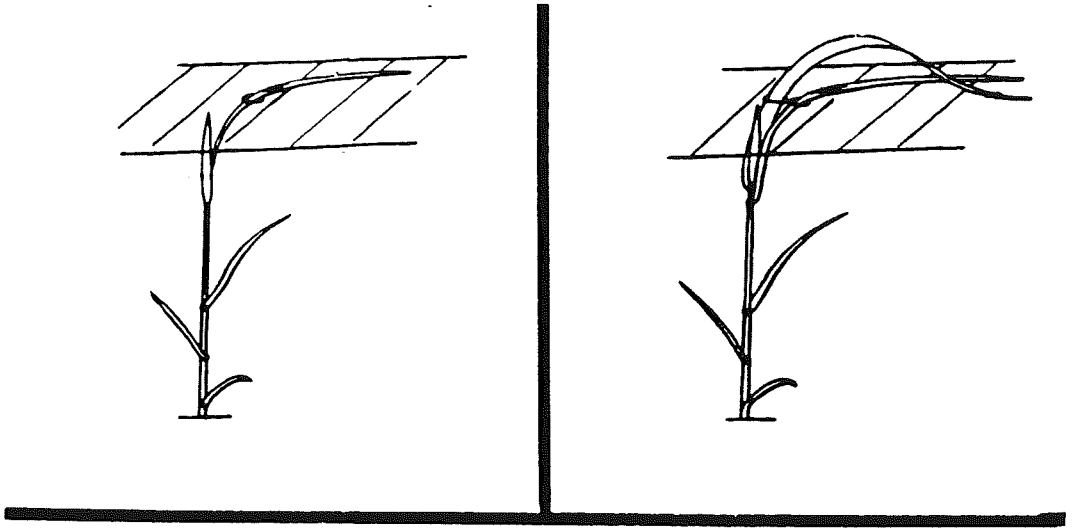


Figure 3. The plant on the left has the 4th true leaf floating while the one on the right has the 5th leaf becoming aerial.

Tillering occurs shortly after the aerial leaf emerges from the water. Figure 4 depicts a plant with one and two tillers. Usually the shallower the water and the less dense the plant population, the more tillers will develop. Tillers develop from the base of the plant in the axils of the leaf blades.



Figure 4. The plant on the left has one tiller, and the right one two tillers. The tillers are depicted in black. The seedlings would be about 12 inches tall.

During the tillering stage the number of kernels on the panicle are also determined, thus stress such as nutrient deficiency during this stage of development can reduce kernel number and tiller number.

The boot and flowering stages are important especially for chemical disease control and nitrogen application if nitrogen deficiency is observed at this stage. The flag leaf is out and needs to be free of disease for optimum yield. Figure 5 depicts the boot and flowering stages.



Figure 5. The plant on the left is in the boot stage, the center one has the female flower completely out and the male flowers partially out, while the right one has the panicle (both female and male flowers) emerged. Pollen is shed when the male flowers have emerged.

The wild rice kernel develops after pollination which occurs shortly after the female flowers emerge from the boot. The hulls (lemma and palea) develop first and the kernel elongates inside (figure 6).

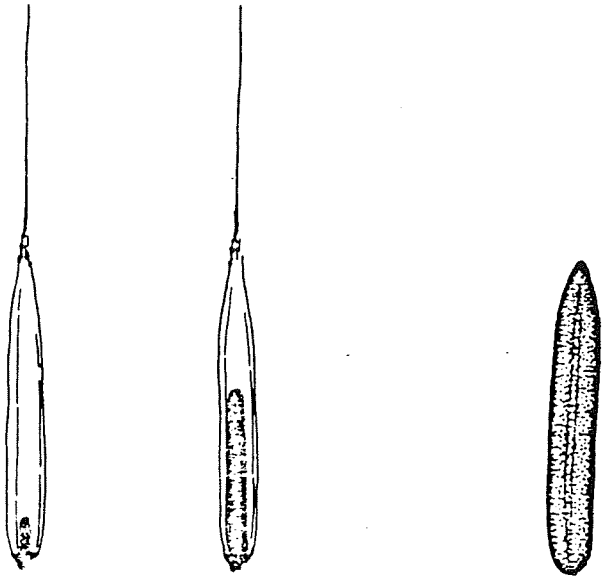


Figure 6. The kernel is just developing inside the hulls in the left grain, while in the center grain it is about half developed. The mature kernel (hulls removed) on the right has a collapsed vein on one side.

Figure 7 depicts a flowering wild rice plant. Plant height usually is about 4 to 7 feet.

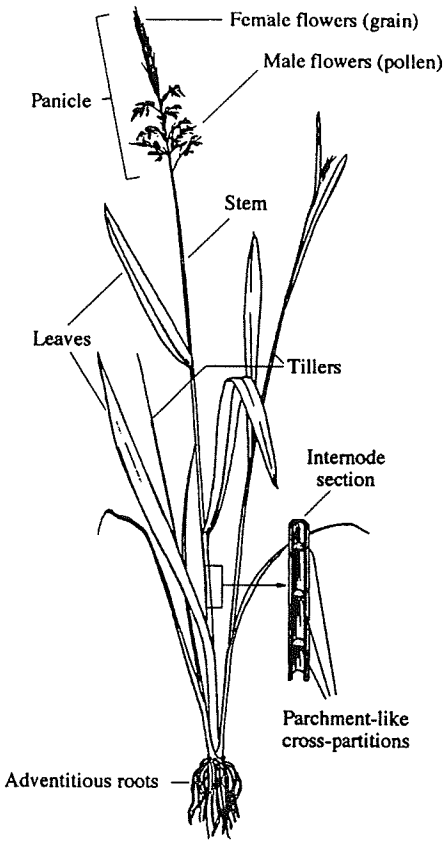


Figure 7. A flowering plant with the female flowers emerging from one tiller, and the second tiller before the boot stage.

The wild rice plant begins its most rapid growth during tillering and jointing (Figure 8). Nutrient availability and lack of stress are important to obtain optimum yield.

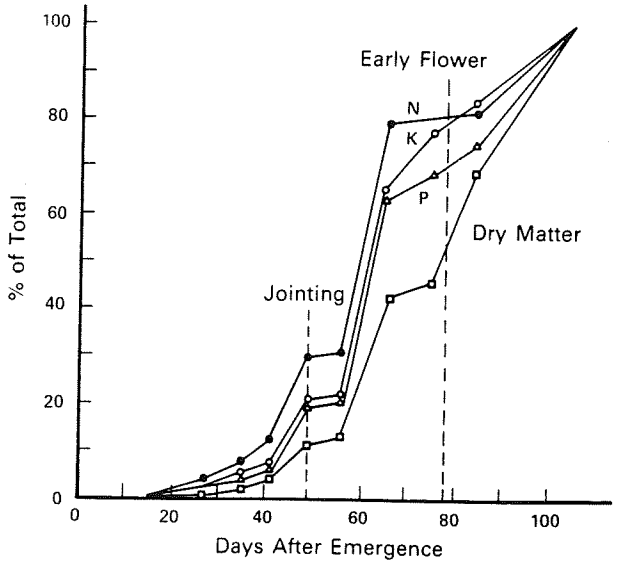


Figure 8. Accumulation of dry matter, nitrogen, phosphorus, and potassium in wild rice during the growing season.

Practices for Nutrient Management

Nitrogen

As with most grains, N management is very important in obtaining high yields. Nitrogen deficiency is a common problem.

Wild rice paddies are very different from upland fields. Nitrate build-up in ground waters due to over fertilization is not a problem. Most N losses occur by the process of denitrification which produces N_2, which makes up 80% of the atmosphere.

Basal N is applied in the fall. To maximize carry over of N to the following spring and maximum fertilizer efficiency, application should be made when soil temperatures are below 50°F, preferably as close to freezing as possible. See Table 1 for rates.

Dry urea, liquid fertilizers or anhydrous ammonia should be incorporated. Nitrate fertilizers are ineffective because of denitrification upon flooding.

Fall flooding is recommended where possible to prevent oxidation of the ammonium forms of N to nitrate with subsequent losses due to denitrification in the spring.

A minimum of one application of about 35 lb/ac of N urea, applied by aircraft, is usually necessary. Additional application may be applied if needed. Drainage of paddy water should not occur for about 5 days after N application.

Field testing in mid-June for soil ammonium-N can be used to determine if topdressing is needed and to determine how much is needed. The SPAD chlorophyll meter is also a useful aid in determining the on-set of N deficiency in wild rice.

Phosphorus

Phosphorus is an important nutrient for plant growth. In well fertilized paddies, P accumulates over many seasons and P deficiency is now uncommon.

Phosphorus is the nutrient most limiting for algae production in fresh waters, and hence, is of great concern in surface water quality. Phosphorus in soils, however, is very insoluble and usually moves into surface waters in particulate form (by soil erosion). However, when phosphorus is surface applied, it can dissolve in paddy waters causing algal blooms. Incorporation of P fertilizer is recommended and erosion from ditches, etc. should be minimized to prevent phosphorus from entering drainage waters.

Phosphorus should be applied in the fall. It must be incorporated by plowdown or injection. Surface application results in elevated P in the flood water and algal blooms. The rate of application should be determined by soil testing. When the Bray P-1 test exceeds 16 ppm, do not apply P. See Table 1.

Potassium

Potassium is required by wild rice both for high yield potential as well as helping in protection against some diseases.

Potassium at the levels applied to wild rice paddies is not an environmental concern.

Potassium is usually applied in the fall with the phosphorus. See Table 1 for rates. Some growers apply potassium with the topdress N. This increases the late season uptake of K and might help prevent some diseases.

Table 1. Recommendations for fall fertilization of wild rice.

Nitrogen	Criteria	Amount to Apply (lb/A)	
		Mineral Soils	Organic Soils
Nitrogen	Status of Paddy:	Incorporated	Incorporated
	First year only	70-100	25-40
	Second year and older	70-100	50-70
Phosphorus	Soil Test Results Bray P-1 (ppm)	Amount to Apply (lb/A)	
(p_2O_5)	0-7	40-50	
	8-15	20-30	
	16 +	0	
Potassium	Soil Test Results (ppm)	Mineral Soils Incorporated	Organic Soils Incorporated
	0-50	80	120
	51-100	50	90
	101-150	30	40
	151 +	0	0

Water Drainage Before Harvest

Nutrients in the water can be detected in wild rice fields for 3 to 5 days after fertilization but at drainage time most nutrients are used by the plants. However, to avoid erosion of drainage ditches the water should be released slowly during a one to two week period before harvest. The soil particles from erosion can carry phosphorus into surface waters. Drainage ditches should be stabilized by grasses if possible.

Practices for Weed Management

Identification

The weeds in wild rice fields are the common water weeds of Minnesota. Broadleaf weeds are more serious than grass weeds. Identification of weeds is important so the proper control measures can be applied.

A description of the most common weeds found in wild rice fields is given below:

Common Waterplantain

Common waterplantain (*Alisma trivale* Pursh) is an emersed, erect, aquatic perennial that reproduces by seed and rootstocks. It is the most troublesome weed in wild rice. It may grow 48 inches tall and has elliptical leaf blades, 10 inches long and 6 inches wide. The leaves have long petioles. Leaves of plants from rootstocks emerge from the water before wild rice to form a dense leaf canopy that shades wild rice. This kills some plants and reduces tillering of wild rice. Seeds germinate during much of the growing season. Seedling waterplantain does not injury wild rice; however, it will produce seed and develop root-stocks. Common waterplantain is more severe in shallow (1-6 inches) water, but will grow in water up to 30 inches deep.

Cattail

Cattails (*Typha latifolia* L.) are perennials that reproduce by rootstocks and by minute airborne seeds that germinate readily in mud and shallow, clear water. Established plants have an extensive rhizome system and erect stems with long, narrow leaves. Cattails grow in areas of the field with poor drainage. Cattails on edges of dikes are overwintering sites for the wild rice stem borer.

Burreed

Burreed (*Sparganium eurycarpum* Engelm) is a perennial that reproduces by seeds and spreads by rhizomes. Plants are 3-4 feet tall and erect with a sparingly-branched, leafy stem. Leaves are narrow and upright. Inflorescence is forked with seed heads, bur-like before seeds fall away. Seeds are pyramidal in shape. Plants occur in shallow water and can spread rapidly.

Common Arrowhead

Common arrowhead (*Saggitaria latifolia* Willd) is a perennial that reproduces by seeds, rhizomes, and tubers. Common arrowhead is a rooted aquatic with arrow-shaped leaves. Lower leaves are basal with petioles as long as the water is deep. Plants are

1-2 feet tall with white flowers. Seeds are flat, 1/8-inch wide with marginal wings, and are clustered into a spherical head. It occurs in shallow water but is not prevalent enough to cause serious problems.

Cursed Crowfoot

Cursed crowfoot (*Ranunculus sceleratus* L.) is an annual or short-lived perennial, mostly emersed, with pale yellow flowers on water surface. Lower leaves have long petioles. Seed heads are cylindrical. It usually occurs in shallow parts of fields where wild rice stands are poor.

Water Starwort

Water starwort (*Callitriche heterophylla* Pursh) is an aquatic herb with narrow and opposite leaves on a stem below the water. Leaves at top of stem float and are arranged in a rosette. Plant reproduces by seeds and can be found in all parts of a field where wild rice stands are poor. It can reduce wild rice stands.

Small Pondweed

Small pondweed (*Potamogeton pusillus* Fern.) is an aquatic perennial that reproduces by seeds and underground buds. Its stems are slender with numerous branches toward the top. Leaves are all submersed, narrow, and with three veins. Flowers in elongated spikes are borne on axillary stalks. Whorls of egg-shaped, beaked seed (achenes) develop in spikes.

Small pondweed can be found in shallow areas of a field where wild rice stands are poor. Often it can be found in interior drainage ditches.

Sago Pondweed

Sago pondweed (*Potamogeton pectinatus* L.) is an aquatic perennial herb that reproduces by seed and tubers. The plant has a bushy appearance because of its much-branched stems and numerous threadlike leaves spreading in a fan-like fashion. Leaves are submerged, very narrow, tapering to sharply pointed tips. This pondweed can grow in deeper water. Adequate wild rice stands will help shade out this weed.

Other Weeds

Other weeds found in wild rice fields are water parsnip (*Sium suave* Walt.), nodding beggarsticks (*Bidens cemua* L.), marsh cress [*Rorippa palustris* var. glabrata (Lunell); Vict.], barnyardgrass (*Echinochloa crusgalli* L.), spikerush (*Eleocharis* spp.), duckweed (*Lemna minor* L.), chara (*Chara* spp.), naiad (*Najas* spp.), American sloughgrass [*Beckmannia syzigachne* (Steud.) Fern.], and reed-meadow grass (*Glyceria grandis* S. Wats.).

Cultural Practices for Control

When planting new fields, weed free wild rice seed should be planted to avoid introduction of weeds. Seed for planting should be harvested from a weed free field especially of the most troublesome weed, common waterplantain.

Good fall tillage of fields can reduce weed problems. Fall moldboard plowing will bury common waterplantain corms deep enough so they can't produce plants the following year. Fall plowing will also help control burreed and cattails. Fall rotovating will help control weeds but it is not as effective as plowing. If common waterplantain persists, fall flooding will help control this weed by killing the corms.

Rotation with an upland crop will help control aquatic weeds. Maintaining a water depth of 10 inches on the shallow end of a field during the growing season will help in controlling weeds. Incorporation of fall fertilizer, especially phosphorus, will reduce the chances for algae in fields the following spring.

Chemical Control

The only chemical available for weed control is 2,4-D amine to which wild rice is not as tolerant as it should be to obtain good common waterplantain control. Only one application at 1/4 pound active ingredient per acre can be applied when wild rice is in the tillering stage. It cannot be applied after wild rice begins jointing since serious injury can result. Spot spraying of areas with heavy infestations of common waterplantain is the best chemical control approach. 2,4-D amine is very soluble in water and has a half-life of 10 days, thus do not drain fields for 10 days after application.

Algae growth can be partially controlled with copper sulfate at 15 pounds per acre. Retreatment is often necessary for better control.

PRACTICES FOR DISEASE MANAGEMENT

Introduction

There are at least nine fungal and two bacterial diseases which affect cultivated wild rice in Minnesota. Fungal brown spot and spot blotch are caused by two different fungi have significant biological and economic impact on cultivated wild rice production in Minnesota.

Fungal and the bacterial disease of wild rice pathogens can also infect many other wild plants and cultivated crops. These other plants offer an opportunity for the wild rice pathogens to sustain and, perhaps, increase themselves in the absence of wild rice. The following is an outline of possible practices to help manage wild rice diseases, particularly fungal brown spot. To be successful, an integrated disease management program must be employed.

Identification of diseases.

Fungal: The most serious disease is fungal brown spot caused by *Bipolaris sp.*. Leaf spots that are oval in shape, often with yellow margins, are the common symptoms.

Bacterial: The most common is bacterial leaf streak caused by the bacterium, *Pseudomonas syringae*. Symptoms are long, narrow, dark green, water-soaked streaks that expand lengthwise along the leaf blade.

Cultural practices for control

Wild rice varieties: All current varieties are susceptible to the fungal and bacterial pathogens which infect wild rice. The use of earlier maturing varieties may result in the less fungal brown spot yield and resulting reductions. Disease occurring at (one-quarter gain fill) or later in wild rice development will not significantly affect yield. Wild rice varieties with more upright leaf architecture will allow rain to run-off allowing for less favorable infection sites.

Plant nutrition: Proper nitrogen fertilization levels and timing will result in healthier wild rice. Nitrogen excesses during early plant growth and/or deficiencies during fertilization may weaken the plant's ability to defend themselves against disease. Removal of paddy water early may interfere with nutrient uptake and thus, affect plant health and disease susceptibility.

Incorporation of residue: Incorporating wild rice crop residue into the soil after harvest will facilitate its decomposition. The loss of this protective residue for the pathogens will greatly reduce their chances of surviving over the winter. The residue should be chopped up into small pieces. Disease organisms are not likely to survive in smaller pieces of residue that are burned in the soil.

Plant population: Wild rice stands of 2-6 plants/ft^2 ensures proper nutrient uptake and healthy plants. This population also allows for quicker drying of the leaf canopy after periods of dew and rain, thus reducing the ability of the fungal brown spot and bacterial pathogens to infect and cause disease.

Crop rotation: Practice rotation cropping or fallow for at least one year between wild rice crops. Fall flooding, burning or incorporating wild rice crop residue helps reduce the over-winter pathogens. Select alternative crops, such as legumes, which will not support wild rice pathogens.

Weed control: Good aquatic weed control will reduce weed competition for space and nutrients. Weeds serve as alternate hosts for several fungal and bacterial pathogens of wild rice.

Dike management: Paddy dike management will help control diseases. Dike grasses and weeds which can serve as hosts for wild rice pathogens should be cut or burned, if possible. Delay mowing dikes when possible until July 15 to enhance wildlife habitat.

Fall flooding: Flooding fields in the fall, if appropriate can help control diseases. Disease organisms do not survive in water for long periods. Thus, reducing the amount of inoculum available for infection in the spring. Fall flooding will reduce weed populations thus reducing competition and wild rice pathogens which can infect and increase on these weeds.

Chemical control

Propiconazole (Tilt^{R}) is the only fungicide available to control fungal brown spot and other fungi on wild rice. Label recommendations should always be followed for spray intervals, rates and timing to give the most effective disease control.

The use of Tilt usually results in good disease management and economic return.

Practices for Insect and Other Pest Management

Identification

Pest identification is important to determine control measures. The most common pests are described below.

Wild Rice Worm

The wild rice worm (*Apamea apamiformis*) in the larval stage is the most widespread insect in wild rice. Adult moths appear in late June and lays up to 150 eggs in the florets. The larvae stages feed on the developing kernels and exit to feed on additional kernels. If not treated, losses of 50% or higher can occur. One larva per panicle will reduce yield by 11%.

Midge

A variety of midges in the families Chironomidae and Dixidae use flooded fields for larvae development. One of the Dixid midges, *cricotopus* spp. can cause severe injury in newly planted fields by feeding on the plants while submerged.

Rice Stalk Borer

The larval stage of the rice stalk borer (*Chilo plejadellus*) feeds on the inner stem tissue. Moths lay eggs toward the end of June on floating leaves. The larvae initially feed on the leaf and then bore into the main stem. Stem breakage, white panicles and poorly filled grains can occur with infestation. The insect overwinters as a larva primarily in wild rice stems, cattails and pigweed.

Tadpole Shrimp

The adults of tadpole shrimp (*Triops coulsi*), which hatch from eggs from the previous years, can be found in the water early in the spring before wild rice emerges and can feed on wild rice seedlings and disturb the soil causing difficulty in seedling establishment. Tadpole shrimp are most often a problem in new fields.

Other Insects

The rice water weevil, rice leaf minor, and rice stem maggot can also be found but usually no economic damage occurs from these.

Other Pests

Crayfish can cause serious losses by foraging on the soil surface resulting in destruction of plants. Blackbirds are a major problem during grain fill.

Cultural Practices

There are no cultural controls for the wild rice worm or midge. Wild rice stem borer can be partially controlled by fall flooding and keeping the fields and dikes free of cattails, pigweed and other plants for winter survival of the larva. Crayfish and blackbirds are also difficult to control by cultural practices. Blackbird control can be partially done by reducing cattail habitats and by shooting with steel shot, propane guns or other noise makers.

Chemical Control

Malathion is cleared for use on wild rice worm and should be applied at one pound active ingredient per acre when the larvae are exposed outside the hull, usually 14 to 21 days after eggs become visible in the hulls. The half-life of malation is one day i.e. one-half of malathion decomposes at 67°F to products other than the original material. It is also readily adsorbed by the soil. Application restrictions are not to apply within 10 days of harvest.

Table 2. Properties of pesticides used in, or associated with, wild rice production.

Trade name	Common name	Soil Sorption[1] Index (KOC)	Solubility[2] (ppm)	Half-life[3] (days)
2,4-D	2,4-D amine	20	796,000	10
MCPA	MCPA amine	20	86,000	25
Malathion	Malathion	1800	130	1
Tilt	Propiconazole	650	110	110
Mancozeb	Mancozeb	2000	6	70

[1] Soil sorption index (KOC) = an index of the degree of adsorption at PH 7.0 of a compound to soil normalized for carbon content; the higher the KOC value, the stronger the compound is adsorbed to the soil.

[2] Solubility = in water at PH 7.0; The properties of some pesticides vary with pH.

[3] Half-life = the time required for one-half of the pesticide to decompose to products other than the original pesticide at 67° F.

Practices for Pesticide Application and Handling.

- Wear proper protection equipment when handling pesticides.
- Read and follow all label directions.
- Comply with all federal and state laws regulating pesticides.
- Keep material safety data sheets accessible.
- Become a certified applicator
- Calibrate application equipment properly.
- Protect water resources from contamination when mixing and loading pesticides and when rinsing equipment and pesticide containers.
- Provide secure, safe storage for pesticide containers.